

LINKIFIC – SYSTEM ARCHITECTURE

Technical Architecture & Component Explanation

1. Architecture Overview

Linkific uses a layered architecture. React provides the interface, FastAPI manages APIs and business logic, authentication and RBAC protect operations, the AI chatbot interprets natural-language requests, and PostgreSQL stores application data.

2. High-Level Architecture

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graph TD
    User --> ReactFrontend[React Frontend]
    ReactFrontend --> FastAPIBackend[FastAPI Backend]
    FastAPIBackend --> AuthRBAC[Authentication / RBAC]
    AuthRBAC --> AIChatbot[AI Chatbot / Tools]
    AIChatbot --> BusinessLogic[Business Logic]
    BusinessLogic --> PostgreSQL[PostgreSQL]
```

3. Layer Responsibilities

- React Frontend – UI, dashboard, tasks and chatbot interaction.
- FastAPI – API endpoints and backend coordination.
- Authentication – identifies the current user.
- RBAC – checks role and permissions.
- AI Chatbot / Tools – interprets natural-language requests.
- Business Logic – validation and HR operations.
- PostgreSQL – persistent application data.

4. RBAC Flow

User Request → Authentication → Role & Permission Check → Allow / Deny → Backend Action

The backend is the final authorization boundary. Even when the chatbot understands a request, the backend independently verifies permission.

5. Role-Based Chatbot Task Assignment

Admin → Chatbot → Identify Task Request → Check Permission → Validate Employee → Create Task → Confirmation

6. Security Considerations

- Do not rely on chatbot instructions as the only security control.
- Validate authentication before protected actions.
- Check role and permission on the backend.
- Validate employee and task data.

- Return controlled authorization and validation errors.

7. Example Data Flow

For “Assign Rahul a task to complete onboarding documentation,” the request reaches the backend, the user's authorization is checked, the employee is validated, business logic creates the task, PostgreSQL stores the data, and confirmation is returned.

8. Future Architecture Improvements

- Resource-level permissions.
- Stronger audit logging.
- Sensitive-action confirmation.
- Monitoring and analytics.
- More modular and independently testable AI tools.
- Expanded HR chatbot capabilities.

9. Architecture Summary

The separation of frontend, backend, authorization, AI, business logic and database responsibilities improves security, maintainability, testing and extensibility.